

The empirical recurrence for OEIS A320405, recovered from its tail

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Abstract

OEIS A320405 records “Empirical recurrence of order 83” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 108$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 86$. Its last coefficient is $c_{83} = 4 \neq 0$, so no further tail satisfies anything shorter; any order-83 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A320405 is “Number of $n \times 4$ 0..1 arrays with every element unequal to 0, 1, 2, 4 or 5 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

8, 35, 156, 786, 3761, 18725, 92578, 463045, ...

The entry states:

Empirical recurrence of order 83 (see link above)

As of the “Last modified” line on the live entry (Oct 12 2018 12:36:17, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 108$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 108$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 83: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 83.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 216$ exact terms from $n = 3$ on, it returns order 83 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{83} c_i a(n-i),$$

c_1	8	c_{22}	−1601784	c_{43}	6176101	c_{64}	−599296
c_2	−12	c_{23}	3831810	c_{44}	−18197550	c_{65}	521260
c_3	−32	c_{24}	415303	c_{45}	5933036	c_{66}	−117830
c_4	120	c_{25}	−4388990	c_{46}	10279549	c_{67}	−80883
c_5	−136	c_{26}	3611887	c_{47}	−12223160	c_{68}	131044
c_6	−590	c_{27}	3408366	c_{48}	640964	c_{69}	−87880
c_7	2336	c_{28}	−7013572	c_{49}	−83975	c_{70}	9671
c_8	−1244	c_{29}	8879180	c_{50}	3798467	c_{71}	43107
c_9	−4990	c_{30}	15826852	c_{51}	1279986	c_{72}	−47577
c_{10}	12266	c_{31}	−6158370	c_{52}	4228588	c_{73}	27778
c_{11}	28	c_{32}	−16131824	c_{53}	−832602	c_{74}	−9195
c_{12}	−43819	c_{33}	−3999600	c_{54}	−4756227	c_{75}	−485
c_{13}	27506	c_{34}	−26342479	c_{55}	694843	c_{76}	2532
c_{14}	44668	c_{35}	−15563274	c_{56}	−160307	c_{77}	−1431
c_{15}	−123622	c_{36}	17697172	c_{57}	2381705	c_{78}	311
c_{16}	17686	c_{37}	15617361	c_{58}	−193635	c_{79}	5
c_{17}	644886	c_{38}	21525479	c_{59}	−1681503	c_{80}	−27
c_{18}	−438579	c_{39}	10239041	c_{60}	537398	c_{81}	25
c_{19}	−570382	c_{40}	7689585	c_{61}	−150598	c_{82}	−16
c_{20}	1391613	c_{41}	−10623931	c_{62}	290523	c_{83}	4
c_{21}	−2256167	c_{42}	−13452711	c_{63}	80236		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 86$, and it is the recurrence OEIS A320405 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{83} - c_1x^{83-1} - \dots - c_{83}$. Its constant term is $-c_{83} = -4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 83 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 83, so $\deg q = 83$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 83, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 83, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-83 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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